



Week 09: Java Web Application Development

Dynamic Web Application using Java Servlets

Objective:

The objective of this lab assignment is to understand Java servlets and their role in web application development. Learn how to create a simple servlet, handle HTTP requests and responses, and deploy the servlet within a web server.

Lab Assignment Steps:

1. Prerequisites:

- Ensure that a servlet container like Apache Tomcat installed.
- Set up the development environment in your preferred Integrated Development Environment (IDE) such as Eclipse or IntelliJ IDEA.

2. Creating a Simple Servlet:

- Create a new Dynamic Web Project in your IDE.
- Create a new Servlet class named "SimpleServlet" that extends HttpServlet.
- Override the doGet() method to handle HTTP GET requests.
- Implement the servlet logic to generate a simple HTML response with a greeting message.

3. Building the Servlet:

- Compile the servlet class to generate the corresponding .class file.
- Ensure that the servlet class is located within the correct directory structure as per the servlet container's requirements.

4. Deploying the Servlet:

- Deploy the servlet project to the servlet container (e.g., Apache Tomcat).
- Start the servlet container and ensure that it is running without errors.

5. Accessing the Servlet:

- Open a web browser and navigate to the URL specified in the servlet mapping.
- Verify that the servlet responds with the expected greeting message.

6. Experimenting with Servlet Functionality:

- Modify the servlet code to accept parameters from the HTTP request using two different methods:
 - 1. Query parameters
 - 2. HTML form data
- Update the servlet response to include dynamic content based on the request parameters.
- Experiment with different HTTP methods (GET, POST, etc.) and handle them accordingly in the servlet code.

7. Troubleshooting and Debugging:

- Debug any errors or issues encountered during servlet development and deployment.
- Use logging statements and debuggers to identify and resolve issues in the servlet code.

8. Reflection:

- Reflect on the challenges faced and lessons learned during the servlet development process.